SIMONE UTILI

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PROFESSIONAL SUMMARY

MSc Robotics student at TU Delft working on robot learning for manipulation and humanoids: VLA fine-tuning, reinforcement learning and real-to-sim digital twins. **Available for a 6+ month MSc thesis from January 2027.**

EDUCATION

Master of Science in Robotics 2025 – Present
Delft University of Technology, Netherlands *Current GPA*: 8.5/10
Key modules: Machine Learning for Robotics, Perception and 3D Mapping, Intelligent Control Systems, Motion Planning, Human-Robot Interaction, Deep Reinforcement Learning

Bachelor of Science in Automation Engineering 2021 – 2025
University of Bologna, Italy *Final Grade*: 110/110 cum laude
Key modules: Analytical Mechanics, Model-Based Control, Adaptive Control, Automated Machines, Electronics

Technical High School Diploma 2016 – 2021
ITIS Nullo Baldini, Ravenna, Italy *Specialization*: Electronics *Final Grade*: 100/100

PROFESSIONAL EXPERIENCE

Embodied AI Engineer May 2026 – Present
Motion1 (Humanoid Robotics Startup) *Brussels, Belgium / Remote*

- VLA training stack for Unitree G1 and AgiBot G2: RL post-training in simulation, and physical and visual sim data augmentation.
- Real2Sim: agentic LLM-driven digital twins (3D Gaussian Splatting, 3D asset generation) and dexterous tasks reproduced from egocentric human video via 4D human–object interaction and closed-loop RL.
- Video annotation for physical AI: SLAM-based hand tracking, object tracking, semantic and action labelling.
- Built the integration layers adapting open-source frameworks to these use cases.

Research Analyst July 2025 – December 2025
Clust-ER Mech *Emilia-Romagna, Italy*

- Co-authored a regional white paper on collaborative-robotics adoption in Emilia-Romagna manufacturing; collected and analysed data from companies, universities and technology centres to benchmark cobot applications, barriers and market trends.

Simulation Robotics Engineer (Intern) March 2024 – June 2024
Laboratory of Automation and Robotics, University of Bologna

- Controlled a robotic manipulator in NVIDIA Isaac Sim for pick-and-place tasks, implementing RMPFlow motion generation and collision avoidance via Omniverse APIs.

TECHNICAL SKILLS

Programming	Python, C/C++, MATLAB/Simulink, Arduino, Git, Docker, LaTeX
Robotics Software	ROS/ROS2, PyBullet, Gazebo, NVIDIA Isaac Sim/Lab
ML Libraries	PyTorch, JAX, TensorFlow, scikit-learn, Stable Baselines3
Models & Frameworks	NVIDIA GR00T, Cosmos, V2D, SONIC, SAM 3D, FoundationPose, FLUX.3-Action
Embodiments	PAL TIAGo, Unitree G1, AgiBot G2 (sim only), UR5e (sim only), SO-101
Languages	Italian (Native), English (C1 - Advanced), French (A2 - Beginner)

PUBLICATIONS

S. Utili, A. Govoni, G. Palli, "Online Real2Sim Calibration of Velocity-Controlled Digital Twins via Reinforcement Learning," in *Proc. 2026 IEEE/ASME Int. Conf. on Advanced Intelligent Mechatronics (AIM)*, Genova, Italy, 2026, pp. 1–6. DOI: 10.1109/AIM65483.2026.11658146

SELECTED PROJECTS

Haptic Feedback for Teleoperation Teaching – HRI Study March 2026
MSc Human-Robot Interaction Course Project (team of 4) *TU Delft*

- Designed a five-condition user study evaluating haptic guidance strategies, including virtual walls, fading assistance, and learned-trajectory feedback, for teleoperation training using a Haply device.
- Built the full pipeline: real-time Gaussian Process trajectory learning from demonstrations and haptic rendering of guidance forces.

Python, PyGame, scikit-learn (GP), Haply API

Barista Mobile Manipulator – Motion Planning in Simulation Dec 2025 – Feb 2026
MSc Robotics Course Project (team of 4) *TU Delft*

- Developed a hierarchical motion planner for a mobile manipulator performing bottle pick-and-place in a PyBullet restaurant environment, reaching a 96% task success rate.
- Combined sampling-based global planning (RRT*, RRT, PRM) with MPC tracking in a sequential base-then-arm strategy for approach, grasp, and placement.

Python, PyBullet

RL-Based Digital Twin of a Robotic Manipulator Aug 2024 – Feb 2025
BSc Thesis Project (basis of AIM 2026 paper) *University of Bologna*

- Built a real-to-sim digital twin of the TIAGo arm in NVIDIA Isaac Lab using RL-based adaptive parameter estimation, achieving an 86% reduction in joint-velocity tracking error.
- Validated the estimated parameters in closed loop using high-fidelity Gazebo simulation under real-time ROS control.

Isaac Sim, ROS, Gazebo, Python

ADDITIONAL EXPERIENCE

Zurich Physical AI Hackathon – Black Forest Labs Track October 2026
DARE Campus, Zurich

- Built **Liminal**, a behavior-tree orchestration layer for long-horizon bimanual manipulation with two FLUX.3-Action policies on an XLeRobot (dual SO-101); LoRA fine-tuning took a slot-insertion skill from 0/16 to 3/5 successes. **Awarded 1st place (BFL track).**

HACK2026 – ETH Robotics Club Hackathon on Humanoids March 2026
ETH Hangar, Zurich

- Selected among 50 roboticists from 500+ applicants; built **OpenHumanoid** on the Unitree G1 EDU in 48h, an agentic AI framework for multimodal humanoid locomotion control (OpenClaw gateway, GR00T Whole-Body Controller). **Awarded 2nd place.**

EuroTech x HKTE Hackathon – AI & Robotics Track June 2026
EuroTech Federation & HK Talent Engage — Munich, Germany

- Built **NurseArm**, an agentic system controlling an SO-101 robot arm to automatically sort patients' daily medication, driven by multimodal inputs such as hand tracking and calendar events.

CERTIFICATIONS

IELTS Academic Certificate – Overall Band Score: 8.0/9.0, British Council February 2025
Lifeguard Certification – Società Nazionale di Salvamento (SNS) June 2019